

MATH 180 Strategies of Problem Solving

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Outline

- 1 SoPS Lecture 11
 - Königsberg bridge problem
 - Math and creativity
 - Wrapping up

Review

Last time

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- we learned the degree sum formula
- we learned Euler's formula for planar graphs
- questions about Sprouts

First-year/Transfer Event



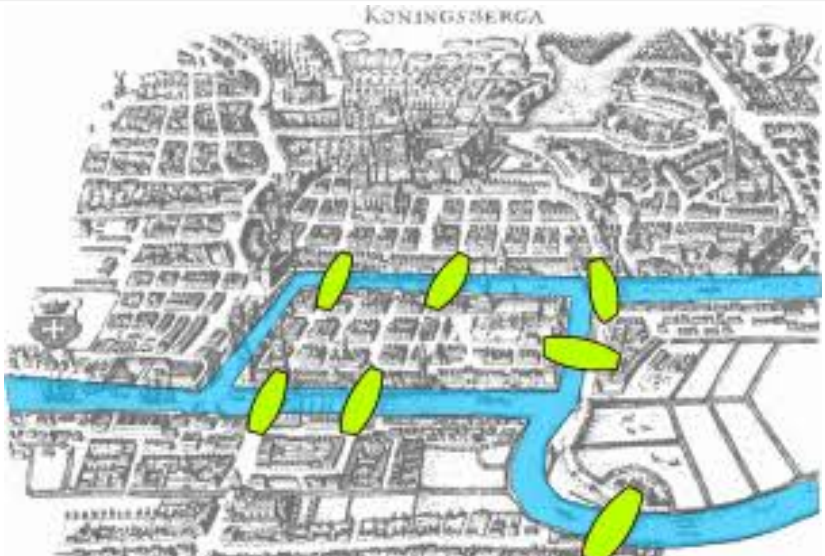
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Bridges of Königsberg



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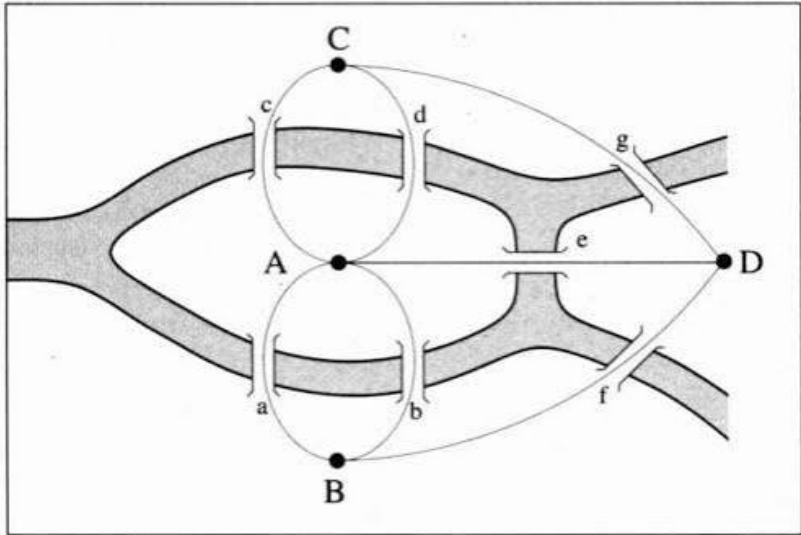
Problem:

Can one find a route which crosses every bridge exactly one time?

Bridges of Königsberg



Bridges of Konigsberg



Eulerian Paths

Defintion

An **Eulerian path** in a graph is a sequence of edges that you can walk along to visit every vertex exactly one time.

Eulerian Paths

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Theorem:

A graph has an Eulerian path if and only if it has less two or less nodes with odd degree.

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Corollary:

The Konigsberg bridge problem has no solution!

Math is creative

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- Euler's Konigsberg bridge problem.

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- $1 + 3 + 5 + 7 + \cdots + 2n + 1 =$

Creative Konigsberg solution



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Challenge problem

Activity:

Color every dot in a 4×4 grid red or blue, avoiding any rectangle with horizontal and vertical sides whose four corners have the same color. Rectangles of every size count. If you succeed, try 5×5 .

Challenge problem

Discussion:

What did you try and what did it reveal?

Computation vs. Creative Math

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Computation:

You're applying a known method. No creativity needed.

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Creative Math:

You're approaching a problem where the way to solve it isn't known. You have to *get creative*.

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- try something deliberately unusual

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- try something deliberately unusual
- change the way data is represented
- examine previous failures and why they failed
- play “what if”
- make a conjecture – then try to break it

Interlude

Activity:

Go to the board and represent the number 12 in different ways, ie. symbols, diagrams, pictures, arrangements, stories.

Promote creativity

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- giving yourself space to work away from everything else
- giving yourself time to work – a beginning and an end
- leveraging the time as long as possible – don't cut thinking short
- being confident – during creation there's no such thing as being wrong
- be playful – tell jokes, try weird things, get silly

Inspirational video

Ministry of Funny Walks (link)

Interlude-revisited

Activity:

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- shares control: "Which direction should we try to investigate next?"

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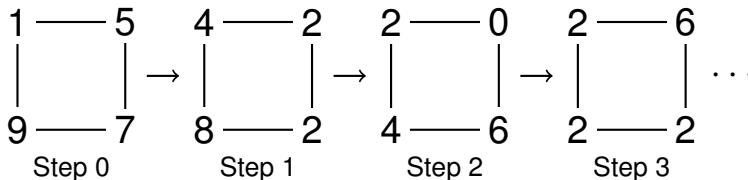
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- makes room for uncertainty: "We can leave that question open for now."
- shares control: "Which direction should we try to investigate next?"
- protects the process of discovery: "would you like a hint, or more time to think?"

New problem

Activity:

Write four nonnegative integers around a square. Replace them simultaneously by the absolute differences between neighboring numbers. Keep repeating. What happens? How long does it take? Can you make something different happen?



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Question of the Day

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Puzzle

Take the numbers $\{1, 2, \dots, 2026\}$. What's the largest subset of them with the property that no number divides another number?

Final thoughts

Reflection

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- Give an example of creativity in mathematics.

Final thoughts

Reflection

- Give an example of creativity in mathematics.
- Give an example of what you can do to help promote creativity.

Submit your responses by **Tonight!**